On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues. ~~Email~~

1)

$$A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 9 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & -3\alpha+9 & 3\alpha-9\gamma & -3\alpha^2+1 \\ 0 & -2\alpha+\beta & 2\gamma+3 & -2\alpha+\alpha^2 \\ 0 & 14-\alpha & -6+\gamma & 2-\alpha \end{bmatrix}$$

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3) $H = I - 2uu^T$

a) $H^2 = HH = (I - 2uu^T) \cdot (I - 2uu^T)$

$= I^2 - 2I(uu^T) - 2(uu^T)I + 4(uu^T)(uu^T)$

$u^T \cdot u = 1$

$H^2 = I^2 - 2I(uu^T) - 2(uu^T)I + 4(u^Tu)u^T$

$= I - 2uu^T - 2uu^T + 4u \cdot 1 \cdot u^T$

$= I - 4uu^T + 4uu^T$

$= I$

b) $H = H^T \rightarrow$ symmectric

$HH = HH^T$

$H^2 = HH^T \qquad H^2 = I$

$HH^T = I \qquad HH^T = H^TH = I$ H is orthonormal.

c) $H = I - 2uu^T$

$Hu = (I - 2uu^T)u$

$Hu = u - 2(uu^T)u$

$Hu = u - 2u(u^Tu)$

$Hu = u - 2u \cdot 1$

$Hu = u - 2u$

$Hu = -u = (-1) \cdot u$ -1 eigenvalue of H.

corresponding eigenvector u.

$A = ADA^{-1} \rightarrow$ digonal

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4) $v(0) = 10$
$w(0) = 40$

$v - w \qquad 10 - 40 = -30$

$v_{k+1} = v_k + 0.30\, w_k$

$w_{k+1} = w_k -$

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